



DESIGN AND DEVELOPMENT OF LOW COST SEMI AUTOMATIC COCONUT LEAF HUSK REMOVAL MACHINE FOR BROOM STICK PRODUCTION

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BONAFIDE CERTIFICATE

Certified that this Make – A – Product (MAP) report “**DESIGN AND DEVELOPMENT OF LOWCOST SEMI AUTOMATIC COCONUT LEAF HUSK REMOVAL MACHINE FOR BROOM STICK PRODUCTION**” is the bonafide work of “**Sabari K (611223114080), Sri Kavın S (611223114101), Naveen Kumar S (611223114064), Tejas T (611223104167) and Ganesh (611223106031)**” who carried out the MAP work under my supervision.

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ABSTRACT

The production of broom sticks from coconut leaves is a widely practiced activity in rural and semi-urban areas, serving as a source of income for many small-scale workers and cottage industries. Traditionally, the process of separating the leaf blade from the midrib (stick) is carried out manually, which is labour-intensive, time-consuming, and inefficient. The manual method often leads to low productivity, inconsistent quality, and physical strain on workers due to repetitive operations.

To overcome these challenges, this project focuses on the design and development of a low-cost semi-automatic coconut leaf husk removal machine that enhances efficiency while remaining affordable and easy to operate. The proposed machine is designed using simple and readily available components such as a wooden frame, a 1/4 horsepower electric motor, a rubber roller, a steel roller, and a blade stripping mechanism.

The working principle is based on a roller–blade system, where the coconut leaf is manually fed into the machine. The rubber roller provides sufficient grip to pull the leaf forward, while the steel roller supports and guides the movement. As the leaf passes through the rollers, a strategically positioned blade removes the leaf blade from the midrib without damaging the stick.

The use of a pulley system helps in reducing the motor speed to an optimal range, ensuring effective operation and preventing damage to the material. The developed prototype significantly reduces manual effort and increases productivity compared to traditional methods. It can process many leaves in a shorter duration with consistent output quality

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CHAPTER 01

BUSINESS IDEA AND PROBLEM STATEMENT (STAGE – I)

1.1 Major Domain

The present project titled “Design and Development of a Low-Cost Semi-Automatic Coconut Leaf Husk Removal Machine for Broom Stick Production” falls under the domain of Mechanical Engineering with a strong interdisciplinary connection to agricultural automation and rural technology development. In many parts of India, especially in coconut-growing regions, the production of broom sticks from coconut leaves is a traditional and economically significant activity. However, the process still relies heavily on manual labor and lacks technological intervention. This project focuses on bridging this gap by introducing a semi-automatic mechanical system that enhances productivity while maintaining affordability. The domain emphasizes the use of simple mechanical components such as rollers, blades, shafts, and motors to achieve efficient material processing. By integrating mechanical design principles with real-world agricultural needs, the project contributes toward sustainable rural development, small-scale industrial growth, and the promotion of low-cost automation technologies that can be easily adopted by local communities without requiring advanced technical expertise.

1.2 Target Customers

The primary target customers for this project include small-scale and medium-scale broom stick manufacturers, rural workers, cottage industries, and self-help groups that are actively involved in coconut leaf processing. These customers are typically located in rural and semi-urban regions where coconut farming is prevalent, and broom stick production serves as a source of livelihood for many families. Due to financial constraints, these users often cannot afford expensive industrial machinery and therefore depend on manual labor for processing coconut leaves. The proposed machine is specifically designed to meet

the needs of such customers by offering a low-cost, efficient, and easy-to-operate solution. Additionally, the machine is suitable for women workers and elderly individuals, as it reduces physical strain and simplifies the process. By addressing the requirements of these user groups, the project aims to improve productivity, reduce labor dependency, and enhance the overall economic condition of small-scale producers.

1.3 Go To Customer (GTC) Process

The Go to Customer (GTC) process for this project involved direct interaction with potential users and stakeholders in the broom stick manufacturing sector. Field visits were conducted to observe the traditional methods used for coconut leaf processing, and detailed discussions were held with workers to understand the challenges they face in their daily operations. Feedback collected during these interactions revealed that the manual process of separating the leaf blade from the midrib is highly labor-intensive, time-consuming, and physically exhausting. Workers also highlighted issues such as inconsistent output quality and low productivity. Based on this information, the project team analyzed the requirements and expectations of customers, which included the need for a simple, durable, and cost-effective machine that can operate with minimal maintenance. The insights gained from the GTC process played a crucial role in defining the design objectives and ensuring that the final product aligns with real-world customer needs.

1.4 Potential Problem Received

During the initial study and customer interaction phase, several potential problems associated with the existing manual method of coconut leaf processing were identified. The most significant issue is the excessive time required to process each leaf, which directly affects overall productivity. Workers are required to manually strip the leaf blade from the midrib using hand tools, which is not only slow but also physically demanding. This repetitive task leads to fatigue, reducing efficiency over time. Additionally, the quality of the output is

inconsistent, as it depends on the skill and experience of the worker. Another major concern is the lack of safety, as manual tools can cause injuries during operation. Furthermore, the dependency on manual labor increases operational costs for small-scale industries, making it difficult for them to scale their production. These problems highlight the urgent need for a mechanized solution that can improve efficiency while maintaining low cost.

1.5 Specific Problem Chosen & Rationale – Customer Fit

Among the various problems identified, the specific problem selected for this project is the manual separation of coconut leaf husk from the midrib, which is a critical step in broom stick production. This problem was chosen because it has a direct impact on productivity, labor efficiency, and product quality. The rationale behind selecting this problem is based on its relevance to a large number of rural workers and small-scale industries, as well as the absence of affordable mechanized solutions in the market. Existing machines, if available, are either too expensive or too complex for rural users. Therefore, developing a low-cost semi-automatic machine ensures a strong customer fit by addressing a real and pressing need. The proposed solution aims to reduce manual effort, increase processing speed, and provide consistent output, thereby improving the overall efficiency of the production process.

1.6 Final Problem Statement

Based on the analysis of customer needs and identified challenges, the final problem statement for this project is formulated as follows: “To design and develop a low-cost, semi-automatic coconut leaf husk removal machine that efficiently separates the leaf blade from the midrib, thereby improving productivity, reducing manual labor, and ensuring consistent output quality for small-scale broom stick production units.” This problem statement clearly defines the objective of the project and provides a focused direction for the design and development process. It emphasizes the key aspects of cost-effectiveness,

efficiency, and user-friendliness, which are essential for successful implementation in real-world applications.

1.7 GTC and Problem – Customer Fit

The alignment between the identified problem and customer requirements was validated through continuous interaction with potential users. Customers expressed a strong need for a machine that is not only efficient but also affordable and easy to operate. The proposed solution meets these expectations by incorporating a simple mechanical design that can be easily understood and maintained by users with minimal technical knowledge. The use of locally available materials and standard components further enhances the practicality of the design. Additionally, the semi-automatic nature of the machine ensures that it provides a balance between manual control and mechanized efficiency. This strong alignment between the problem and customer needs confirms the viability of the project and its potential for successful adoption in the target market.

CHAPTER 02

DESIGN AND DEVELOPMENT OF SOLUTION (STAGE – II)

2.1 Solution Process / Methodology

The solution methodology adopted for this project involves a systematic approach to design, development, and testing of the coconut leaf husk removal machine. Initially, various concepts were explored to identify the most suitable mechanism for separating the leaf blade from the midrib. After careful evaluation, a roller and blade-based mechanism was selected due to its simplicity, effectiveness, and low cost. The design process included the preparation of conceptual sketches followed by detailed CAD modeling to visualize the structure and arrangement of components. Material selection was carried out based on factors such as strength, availability, cost, and ease of fabrication. The fabrication process involved constructing a wooden frame, installing rollers, mounting the motor, and fixing the blade mechanism. Once the assembly was completed, the machine was tested under different operating conditions to evaluate its performance. Necessary modifications were made based on test results to improve efficiency and reliability. This systematic methodology ensured the successful development of a functional prototype.

2.2 Minimum Viable Product (MVP)

The Minimum Viable Product (MVP) developed in this project represents a basic yet functional version of the coconut leaf husk removal machine that demonstrates the core working principle. The MVP consists of a wooden frame that provides structural support to all components, a 1/4 horsepower electric motor that serves as the power source, a rubber roller that provides grip to pull the coconut leaf, and a steel roller that supports and guides the movement of the leaf. A blade is positioned strategically to strip the leaf blade from the midrib as it passes through the rollers. Additionally, a pulley and belt system is used to

reduce the motor speed to a suitable level for safe and effective operation. The MVP focuses on achieving the primary objective of separating the leaf husk efficiently while maintaining simplicity and low cost. It serves as a proof of concept and provides a foundation for further improvements and enhancements.

2.3 Solution – Customer Fit

The developed solution effectively addresses the needs and expectations of the target customers by providing a practical and affordable alternative to manual processing. The machine is designed to be compact and portable, making it suitable for use in rural environments where space and resources may be limited. Its simple construction and operation ensure that it can be used by individuals with minimal technical knowledge. The reduction in manual effort not only improves productivity but also reduces physical strain on workers, thereby enhancing their overall well-being. Furthermore, the consistent output quality achieved by the machine adds value to the final product, making it more competitive in the market. By meeting these requirements, the solution demonstrates a strong customer fit and has the potential for widespread adoption.

2.4 Concept MVP

The concept of the MVP is based on a straightforward yet effective working principle that combines mechanical feeding and stripping mechanisms. In this system, the coconut leaf is manually inserted into the feeding section of the machine, where it is gripped by a rubber roller. The steel roller, positioned opposite to the rubber roller, provides support and ensures smooth feeding of the leaf. As the rollers rotate, the leaf is pulled forward and passes through a blade that is fixed at an inclined angle. The blade removes the leaf blade while allowing the midrib to pass through without damage. The separated materials are collected at different outlets, ensuring clean and efficient operation. This concept emphasizes simplicity, reliability, and ease of use, making it ideal for small-scale applications.

PROTOTYPE DEVELOPMENT (STAGE – III)

3.1 Prototype Development

The prototype of the low-cost semi-automatic coconut leaf husk removal machine was developed using a practical and cost-effective approach, focusing on simplicity, functionality, and ease of fabrication. The structural frame of the machine was constructed using high-quality wood, which was selected due to its availability, lightweight nature, and low cost. The frame was designed to provide adequate support and stability for all mechanical components while ensuring portability. The roller mechanism, which forms the core of the machine, consists of a rubber-coated roller and a steel roller mounted on parallel shafts supported by bearings to ensure smooth and efficient rotation. The rubber roller was specifically chosen to provide sufficient friction for gripping the coconut leaf, while the steel roller acts as a support and guiding element to maintain alignment during operation. A 1/4 horsepower electric motor was used as the power source, and its high rotational speed was reduced using a pulley and belt drive system to achieve an optimal working speed suitable for safe and efficient operation. The blade mechanism was fixed at a carefully selected inclined angle to ensure effective stripping of the leaf blade from the midrib without causing damage to the stick. Proper alignment and spacing between the rollers and blade were maintained during assembly to ensure consistent performance. The entire system was assembled with precision, ensuring that all components functioned harmoniously. The prototype successfully demonstrated the working principle of the machine and served as a functional model for testing and further improvements.

3.2 GTC Iteration & Product Improvement

After the initial development of the prototype, extensive testing was carried out to evaluate its performance under real operating conditions. During this phase, valuable feedback was collected from users and stakeholders through the Go To Customer (GTC) process. Several practical issues were identified, such as slipping of the coconut leaves during feeding, incomplete removal of the leaf blade, and minor misalignment in the roller mechanism. Based on these observations, necessary modifications and improvements were implemented to enhance the overall performance of the machine. The grip of the rubber roller was improved by selecting a better surface finish, and the pressure between the rollers was adjusted to ensure proper feeding of the leaf. The angle and position of the blade were fine-tuned to achieve more effective stripping without damaging the midrib. Additionally, minor structural adjustments were made to improve the stability and durability of the machine. These iterative improvements significantly enhanced the efficiency, reliability, and user-friendliness of the prototype. The process of continuous testing and refinement played a crucial role in optimizing the design and ensuring that the final product meets customer expectations.

3.3 Product – Customer Fit

The final prototype of the machine demonstrates a strong alignment with customer requirements and expectations. It effectively addresses the key challenges identified during the initial study, such as high labor effort, low productivity, and inconsistent output quality. The machine significantly reduces the physical strain on workers by automating the leaf stripping process, allowing them to operate the system with minimal effort. Its compact and portable design makes it suitable for use in rural and small-scale industrial environments where space and resources may be limited. The simplicity of the design ensures that the machine can be easily operated and maintained without requiring specialized technical skills. Furthermore, the consistent output quality achieved by the machine enhances the overall value of the final product. The affordability of the machine makes it accessible to a wide range of users, thereby increasing its

potential for adoption. Overall, the prototype successfully meets the needs of the target customers and demonstrates its practical applicability in real-world scenarios.

BUSINESS MODEL DEVELOPMENT (STAGE – IV)

4.1 Lean Business Model Canvas (BMC)

The Lean Business Model Canvas for this project highlights the key elements required for the successful commercialization of the coconut leaf husk removal machine. The value proposition of the product lies in its ability to provide a low-cost, efficient, and semi-automatic solution for broom stick production. The primary customer segments include small-scale industries, rural workers, and self-help groups engaged in coconut leaf processing. The key activities involve manufacturing, marketing, and distribution of the machine, while key resources include raw materials, fabrication tools, and skilled labor. Revenue is generated through direct sales of the machine, and cost factors include material procurement, labor, and transportation. Distribution channels may include local dealers, agricultural equipment suppliers, and direct sales through demonstrations. Customer relationships can be strengthened through after-sales support and maintenance services. This business model ensures a sustainable and scalable approach for bringing the product to the market.

4.2 Market Potential

The market potential for the proposed machine is significant, particularly in regions where coconut cultivation is widespread. In states like Tamil Nadu, Kerala, and other parts of India, a large number of small-scale industries are involved in broom stick production, creating a strong demand for efficient processing equipment. Currently, most of these units rely on manual labor, which limits their productivity and increases operational costs. The introduction of a low-cost semi-automatic machine provides an opportunity to transform this traditional industry by improving efficiency and reducing labor dependency. The affordability and simplicity of the machine make it highly suitable for rural markets, where cost is a critical factor. Additionally, the growing focus on rural

development and mechanization further enhances the market potential of the product. With proper marketing and distribution strategies, the machine has the potential to achieve widespread adoption and generate substantial demand.

4.3 Business Plan and Marketing Plan

The business plan for this project focuses on the production and distribution of the machine at an affordable cost while ensuring quality and reliability. The manufacturing process can be carried out using locally available materials and fabrication facilities, reducing production costs. The marketing strategy involves creating awareness about the product through live demonstrations, exhibitions, and direct interaction with potential customers in rural areas. Social media platforms and digital marketing can also be utilized to reach a wider audience. Collaborations with agricultural organizations and government schemes can further support the promotion and adoption of the machine. Providing attractive pricing, easy financing options, and reliable after-sales service will help in building customer trust and increasing sales. The overall business plan aims to establish a strong presence in the rural machinery market and achieve long-term sustainability.

4.4 Business – Customer Fit

The proposed product demonstrates an excellent fit between business objectives and customer needs. From a business perspective, the machine offers a viable opportunity for manufacturing and sales due to its low production cost and high demand. From the customer's perspective, it provides a practical solution that improves efficiency, reduces labor effort, and ensures consistent output quality. The alignment between affordability, usability, and performance makes the product highly attractive to the target market. This strong business–customer fit ensures that the product can achieve both commercial success and customer satisfaction.

4.5 Patenting Potential and Startup Potential

The design and concept of the coconut leaf husk removal machine offer potential opportunities for innovation and intellectual property protection. With further refinement and addition of unique features, the machine can be considered for patenting. The simplicity and effectiveness of the design also make it suitable for commercialization through a startup venture. There is significant scope for developing a business focused on manufacturing and supplying low-cost agricultural machinery to rural markets. With the increasing emphasis on entrepreneurship and innovation, this project has the potential to evolve into a successful startup that contributes to rural industrial development.

4.6 SDG Alignment

The project aligns with several Sustainable Development Goals (SDGs), particularly those related to industry, innovation, and economic growth. By providing a low-cost mechanized solution, the project promotes industrial development in rural areas and supports small-scale industries. It also contributes to decent work and economic growth by reducing labor intensity and improving working conditions. Additionally, the project encourages innovation by introducing a practical solution to a traditional problem, thereby supporting sustainable development.

4.7 Unique Selling Proposition (USP)

The unique selling proposition of the machine lies in its combination of affordability, simplicity, and efficiency. Unlike expensive industrial machines, this system is designed specifically for small-scale users, making it accessible to a wider audience. Its compact and portable design allows for easy transportation and installation. The use of simple mechanical components ensures low maintenance and long service life. Additionally, the machine provides consistent output quality, which enhances the overall value of the product. These features

collectively make the machine a highly attractive and competitive solution in the market.

CHAPTER 05

SUMMARY (STAGE – V)

5.1 Minimum Viable Product (MVP)

The Minimum Viable Product developed in this project successfully demonstrates the core functionality of the coconut leaf husk removal machine. It provides an effective solution for separating the leaf blade from the midrib using a simple mechanical system. The MVP serves as a proof of concept and forms the foundation for further development and improvement.

5.2 USP (Technical Dimension)

From a technical perspective, the machine stands out due to its efficient roller and blade mechanism, which ensures effective stripping of coconut leaves. The use of a controlled speed system and proper alignment of components enhances performance and reliability.

5.3 USP (Business Dimension)

From a business standpoint, the machine offers a cost-effective solution that meets the needs of small-scale industries. Its affordability and ease of use make it highly suitable for rural markets, providing a competitive advantage.

5.4 Innovativeness & IP Potential

The project demonstrates innovation by introducing a semi-automatic solution to a traditionally manual process. With further development, the design can be enhanced and protected through intellectual property rights.

5.5 Scope for Startup Creations

The project has strong potential for startup creation, particularly in the field of rural machinery and agricultural automation. With proper investment and marketing, it can be developed into a successful business venture.

5.6 Final Minimum Viable Product (MVP)

The final MVP represents a fully functional prototype that meets the design objectives and customer requirements. It demonstrates the feasibility of the concept and its potential for real-world application.

CHAPTER 06

KEY LEARNING OF THE TEAM (STAGE – VI)

The project provided valuable learning experience in various aspects of engineering and project management. The team gained practical knowledge in machine design, material selection, and fabrication techniques. It also enhanced problem-solving skills by addressing real-world challenges and implementing effective solutions. Team collaboration and communication played a crucial role in completing the project successfully. The importance of field study and understanding customer requirements was clearly realized during the project. Additionally, the team developed skills in testing, analysis, and optimization of mechanical systems. Time management and resource utilization were also key learnings. Overall, the project helped in developing a strong foundation in engineering design and innovation.

CHAPTER 07

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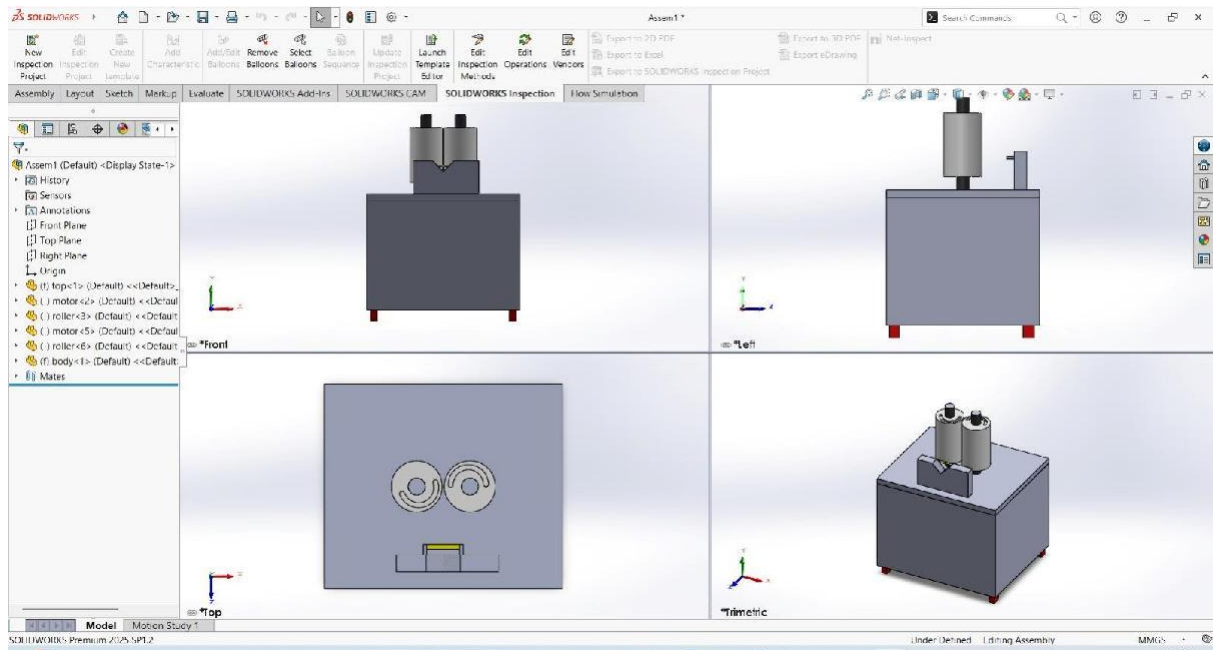


Fig 1 - CAD Model



Fig 2 – Front View



Fig 3 – Top View



Fig 4 – Right Side View



Fig 5 – Left Side View